MD NAIEM MOBASSHIR MONIM

Ahsanullah Hall, BUET
+880 1775-340641 | naiemmobasshir10954@gmail.com
LinkedIn: Md Naiem Mobasshir Monim
Portfolio: www.mobasshirmonim.xyz

Summary

Aspiring engineer with a unique blend of technical acumen and executive leadership experience. Proven track record of managing large-scale initiatives and leading diverse teams while maintaining academic excellence in mechanical systems and computational modeling. Seeking to apply strategic vision and analytical problem-solving to deliver high-impact engineering solutions.

Education

Bachelor of Science in Mechanical Engineering (ongoing, Level 4 Term 2) / *Bangladesh University of Engineering and Technology (BUET)*

Higher Secondary Certificate (HSC) / *Rajshahi College* *2020*

- GPA: 5.00 (Board Scholarship)

Work Experience

Industrial Trainee / *Synovia Pharmaceuticals PLC* *March 2025 – April 2025*

- **Analyzed 8M+** kWh energy consumption data; identified **5.1%** YoY increase and **46%** cost rise; recommended solar adoption and heat recovery systems.
- **Proposed** IoT sensors for predictive maintenance on pharma machinery, estimating **15–20%** downtime reduction via real-time monitoring.
- **Recommended** AI-driven quality control for packaging and LIMS integration, projecting **30–40%** better defect detection and improved traceability across **1000+** test batches/year.
- **Evaluated** HVAC & cleanroom data (temperature, humidity, particle count) across **20+** zones; tuned parameters to maintain >**98%** GMP compliance and cut deviation incidents by 15%.
- **Studied** procurement & inventory in Oracle ERP across **5000+** SKUs; flagged opportunities for data-driven forecasting and supply chain optimization.
- **Collaborated** with **5+** cross-functional teams (engineering, QA, procurement, utilities, IT) on optimizations in water treatment, energy, and production efficiency.

Skills

Programming & Scripting: Python, MATLAB, SQL, C, C++, JavaScript, Kotlin
Data Analysis & ML: Pandas, NumPy, scikit-learn, TensorFlow, Keras, Matplotlib
Data Handling & Visualization: Excel, Google Sheets
Tools & Platforms: Jupyter, Google Colab, Git, Anaconda
Simulation & Engineering: SolidWorks, ANSYS, Simulink, AutoCAD
Other: LaTeX, Bash, Linux
Graphics: Adobe After Effects, Adobe Premiere Pro, Blender, Audacity, Illustrator, Photoshop, Canva
Soft Skills: Communication, Leadership, Research, Teaching, Critical Thinking, Management, Networking, Teamwork

Extracurricular Activities

President / *BUET Entrepreneurship Development Club* *Mar 2025 – Present*

- Spearheaded the university's largest "Entrepreneurs Summit," hosting **50+ stalls** for student startups and corporate entities.
- Orchestrated **7 flagship events** catering to a cumulative audience of **100,000+** across online and offline platforms.
- Secured partnerships with **100+ sponsors** and managed a total financial portfolio of over **50 Lakh BDT**.
- Rose through the ranks from Secretary to President, nurturing **7+ startups** and directing creative branding strategies.

Campus Lead / *Grameenphone* *Jan 2024 – Mar 2025*

- Organized flagship events (Platform SHE roadshow, Next Business Leader) at BUET, aligning club activities with corporate culture.

- Facilitated exposure for **500+** students to Grameenphone's professional ecosystem and events.

Campus Spark / *Unilever Bangladesh* *May 2025 – Present*
- Led brand engagement on campus, increasing participation in UFLP and case competitions.
- Coordinated club collaborations and reported student insights to support recruitment and marketing strategies.

Campus Ambassador / *Perfetti Van Melle Bangladesh* *Jan 2024 – Mar 2025*
- Bridged PVM marketing initiatives with BUET campus activities to enhance brand visibility.

Co-Leader | Animation Lead | Software Sub-team / *Team Interplanetar-A (BUET Mars Rover Team)* *Jan 2023 – Present*
- Contributed to rover communication and perception systems for ERC 2023/2024, ARC 2023/2024, and URC 2023.
- Animated and simulated rover performance for ERC 2023–2025 and IRDC 2023–2024.

General Secretary / *BUET Animata – Animation & VFX Club* *Mar 2025 – Present*
- Served as Founding Assistant General Secretary; organized workshops on basic animation and 3D modeling.
- Designed and animated the official club logo and visual assets.

Association Representative / *Mechanical Engineering Association, BUET* *July 2022 – Present*
- Organized ME FEST 2023, 2024 and the ME Reception 2023.

Projects

Computational Fluid Dynamics with PINNs (DeepXDE)
- Solved inverse 3D Navier-Stokes problems by implementing PINNs in TensorFlow, achieving faster simulation times than ANSYS Fluent while maintaining flow accuracy.
- Reconstructed flow fields in complex geometries, validating the physics-informed approach as a viable alternative to traditional solvers.

Robotics Team – Real-Time Control & Automation / *Team involvement* *Ongoing*
- Engineered autonomous navigation systems using ROS and Raspberry Pi, optimizing sensor fusion and electromechanical integration.
- Deployed ML-based decision algorithms on embedded systems (Arduino) to enhance task planning and control precision.

Time-Series Analysis with RNNs/LSTMs / *Heat Exchanger Monitoring*
- Developed LSTM surrogates to predict thermal performance, significantly reducing the computational cost of repeated ANSYS Fluent simulations.
- Validated deep learning predictions against empirical data and 3D CFD models for intelligent real-time monitoring.

Visual Vocal: Voice-Controlled Robotic Hand / *Real-time Translation*
- Prototyped a speech-to-sign robotic system using Vosk (offline ASR) and Arduino, improving accessibility for the hearing impaired.
- Implemented MediaPipe for gesture validation and precise servo mapping to ensure accurate real-time translation.

Achievements

Entrepreneurship & Leadership Programs
- Champion of Orange Corners BD Ideation Challenge 5.0: Team MycoMatters
- Champion of Cohort 5 of University Innovation Hub Program(UIHP) at RISE, BUET organized by Bangladesh Hi-tech Park Authority and ICT Division.
- Secured 70% scholarship at the ISCEA Prize Global Case Competition 2025
- Finalist – HULT Prize BUET 2026
- Runners Up – Video Essay Challenge, BUET Brainiacs

Rover Competitions:Software
- 1st Place – Anatolian Rover Challenge 2023 (Preliminary Round)
- 2nd Place – Anatolian Rover Challenge 2024 (Preliminary Round)
- 13th – European Rover Challenge 2023 (Remote Edition)
- 15th – International Rover Design Challenge 2023
- 27th – University Rover Challenge 2023

Reference

Dr. Mohammad Mamun
Professor
Dept. of Mechanical Engineering, BUET
Email: dmmamun@me.buet.ac.bd

Dr. Taufiq Hasan
Professor
Dept. of Biomedical Engineering, BUET
Email: taufiq@bme.buet.ac.bd